

Immunological rejection of heart transplant: how lytic granules from cytotoxic T lymphocytes damage guinea pig ventricular myocytes

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Abstract. We investigated the mechanism by which lytic granules extracted from cytotoxic T lymphocytes (CTL) damage guinea pig ventricular myocytes in order to determine whether their actions can be related to the overall immunological rejection of the transplanted heart. Granule-induced myocyte morphological changes and final destruction were preceded by shortening of action potential duration (APD) and reductions of the resting potential and the action potential amplitude. APD shortening was probably caused by a granule-induced increase in outward current (most likely non-specific). Ryanodine, which blocks Ca^{2+} release from the sarcoplasmic reticulum, did not interfere with the morphological and electrophysiological effects of lytic granules. Fura-2 imaging indicated that $[\text{Ca}^{2+}]_i$ initially increased about 2-fold from 90.0 ± 11.5 nM, while cell length decreased less than 5% from a mean value of 99.0 ± 9.0 μm . A further increase in $[\text{Ca}^{2+}]_i$ (>10 fold) was associated with progressive contracture and destruction, suggesting that the structural damage inflicted by lytic granules is caused by $[\text{Ca}^{2+}]_i$ overload. The results indicate that the cytotoxic action of CTL-derived lytic granules may be involved in immunologically induced damage, even to the extent of rejection of the transplanted heart.

Key words: Cytotoxic T lymphocytes – Lytic granules – Transplantation-graft rejection – Myocardial electrophysiology – Action potentials – Membrane currents

Introduction

Heart transplantation has become a common surgical procedure for the treatment of intractable heart failure; the major associated problem is the immunological rejection

induced mainly by cytotoxic T lymphocytes (CTL) [5]. CTL (both CD4 and CD8 cells as well as T helper cells) constitute the major immune effectors responsible for rejection. These cells infiltrate the transplanted heart and destroy the myocardial tissue and endothelium by as yet undetermined cytotoxic mechanisms [12, 16, 17, 19]. A large body of evidence indicates that T lymphocytes are the primary mediators of graft rejection, although it is likely that the Th subtype(s) (CD4), which is involved in delayed-type hypersensitivity (DTH), is at least partly responsible for tissue damage and graft rejection. Nonetheless, although the precise mechanism(s) whereby CTL destroy their targets is of considerable importance and interest, the issue is still highly controversial. It has frequently been suggested that CTL-derived lytic granules, containing the pore-forming protein perforin, are responsible for lysis of the target and that this is the cytotoxic mechanism [6, 13, 23, 31]. Since it has been repeatedly demonstrated that various CTL (and natural killer cells) contain lysosomal lytic granules, which are capable of exerting cytotoxic effects on a wide variety of target cells, we investigated the effects of CTL-derived lytic granules on the morphology, electrophysiology and intracellular calcium concentration $[\text{Ca}^{2+}]_i$ of guinea pig ventricular myocytes. We recorded transmembrane action potentials and membrane currents by means of the whole-cell recording technique, and $[\text{Ca}^{2+}]_i$ using the fura-2 imaging technique. In the present study we show that lytic granules extracted and purified from CTL cause destruction of ventricular myocytes, and that cell death is preceded by electrophysiological derangements and marked increase in $[\text{Ca}^{2+}]_i$. These findings therefore point to a possible mechanism whereby CTL infiltrating the transplanted rejected heart damage the myocardial tissue.

Materials and Methods

Preparation of isolated ventricular myocytes. Adult guinea pig ventricular myocytes were obtained by enzymatic dissociation according to procedures described elsewhere [14]. Guinea pigs (300–400 g) were anaesthetized with pentobarbitone sodium 30 mg/kg i.p. The chest was

opened, the heart rapidly removed and the aorta cannulated and perfused with modified Tyrode's solution in a Langendorff apparatus. After washing out the blood, the heart was perfused with Ca^{2+} -free Tyrode's solution (CFT). Subsequently, CFT solution containing 0.08% collagenase (type 1, Sigma, St. Louis, MO) was circulated for 35–40 min. Thereafter, the heart was perfused with 50 ml of Kraftbrühe (KB) medium, containing high K^+ and low Cl^- concentrations. All solutions were oxygenated and warmed to $36.0 \pm 0.2^\circ\text{C}$. Finally, the heart was removed from the cannula, cut into 3–4 sections and stored in KB medium at room temperature prior to the experiment. The modified Tyrode's solution contained (mM): NaCl, 140; KCl, 4; CaCl_2 , 1.8; MgCl_2 , 1; glucose, 10, and HEPES, 5. CFT solution was prepared by omitting Ca^{2+} from the solution. The KB medium contained (mM): KCl, 70; K_2HPO_4 , 30; MgSO_4 , 5; CaCl_2 , 0.12; glucose, 20; taurine, 20; succinic acid, 5; pyruvic acid, 5; creatine, 5, sodium adenosine triphosphate (Na_2ATP), 5; and ethylene glycol-bis(β -aminoethyl ether) tetraacetic acid (EGTA) 0.5.

Electrophysiological recordings. After an incubation period in the KB medium, a small volume of KB containing myocytes was transferred to the recording chamber (0.5 ml), mounted on the stage of an inverted microscope (IM, Zeiss, Oberkochen, FRG). The Tyrode's solution was superfused at a rate of 1–2 ml/min at room temperature ($24\text{--}25^\circ\text{C}$). For the electrical measurements we used rod-shaped myocytes with cross-striations and clear surfaces. The whole-cell recording technique [10] was used to monitor transmembrane potentials and membrane currents by means of the EPC-7 patch clamp amplifier (List Medical Electronics, Darmstadt, FRG). Electrodes were prepared from soft glass (Kimble, Toledo, Ohio) and had a tip resistance of 1–3 M Ω when filled with the pipette solution. The pipette solution contained (mM): K-aspartate, 120; KCl, 20; MgCl_2 , 3.5; KH_2PO_4 , 20; Na_2ATP , 3; glucose, 10, and EGTA, 1. Measurements were initially carried out when Tyrode's solution (or drug containing Tyrode's) was superfused. Thereafter, flow was stopped, additional measurements made, and 5–7 μl of granule extract added to the bath in the vicinity of the electrode tip. There were no differences in action potential characteristics during flow and after the flow to the bath was stopped.

Preparation of lytic granules. Lytic granules were prepared following methods described elsewhere [2, 20, 24]. In brief, granules were extracted and purified from interleukin-2 (IL-2)-dependent CTL (CTLL-2) cells grown in vitro. This IL-2-dependent cell line is derived from formerly cytotoxic CTL. The CTLL-2 cells were grown in vitro in RPMI-FCS medium supplemented with concanavalin A (Con A) (22%) as an IL-2 source.

Fura-2 imaging. Guinea pig ventricular myocytes were allowed to settle onto the single-thickness glass cover slip bottom of a superfusion chamber. The cells were loaded with 1–2 μM fura-2/AM for 15 min at room temperature without superfusion, then superfused with Tyrode's solution (1–2 ml/min) until the addition of the granules (at least 20 min). Light from xenon lamp was passed through two monochromators (Photon Technology International, Princeton, NJ), to provide the 340 and 380 nm excitation into the epi-fluorescence port of the inverted microscope which was equipped with a 40 $\times$ oil immersion objective (Neofluor; Zeiss, Oberkochen, FRG). The emitted light passed through a 10 nm barrier filter centred at 510 nm and into a CCD camera coupled to a microchannel intensifier plate (Videoscope Technology International, Washington, D.C.). The video signal was captured and analysed by hardware and software in an 80386 compatible computer (Photon Technology International, Princeton, NJ). Four sequential frames were averaged at each wavelength, with approximately four frames lost during transition (full cycle-time 2–3 Hz). At selected times the average frames were stored on hard disk. After the experiment these digitized images were recalled, the 340 nm image was smoothed using a 3 $\times$ 3 convolution algorithm, and the ratio frame created. This image was then masked against the 340 nm image to create a dark background. Because of reported differences between the characteristics of fura-2 free acid and the deesterified fura-2/AM [9], we used an in vivo calibration method. A separate sample of cells were loaded with fura-2/AM in the presence of an EGTA-buffered low Ca^{2+} (10^{-7} M), high K^+ (100 mM) "in-

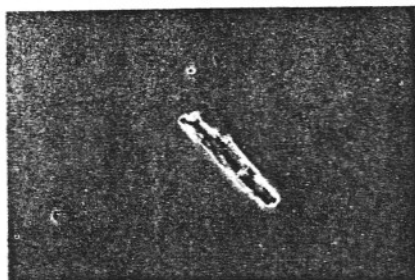
tracellular" solution. The cells were then permeabilized with 250 units of alpha toxin (Gibco BRL, Gaithersburg, MD) for 45 min. Thereafter, a single cell was selected and fluorescence at the two wavelengths measured during exposure to extremes of low and high extracellular Ca^{2+} . The values obtained were: $R_{\min}$ 0.25; $R_{\max}$ 1.2.

Results

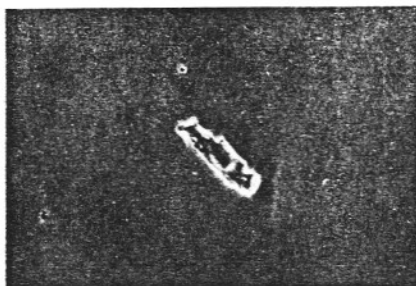
Experiments were carried out by exposing isolated myocytes to lytic granules obtained from the CTLL-2 cells. The exposure to granules resulted in electrophysiological and morphological changes ultimately leading to myocyte contracture and destruction. The typical morphological changes occurring within 5–15 min are illustrated in Fig. 1. In this myocyte (affected quite rapidly by the granules) the most consistent change was a progressive contracture terminating in cell destruction (i.e. cell "death"). The changes in the rod-shape morphology were frequently associated with spontaneous irregular contractions which appeared to originate from different sites within the myocyte. These qualitative observations indicated to us that lytic granules are "toxic" to myocardial tissue and, in principle, can account for the overall outcome of immunological rejection.

The next step was to investigate the effects of lytic granules on electrophysiological properties of guinea pig ventricular myocytes. The aim of these experiments was not only to describe for the first time the effects of lytic granules on myocyte electrophysiology (which might be relevant to CTL cytotoxic action); we thought that any granule-induced changes in myocyte electrophysiology would help to elucidate the mechanisms involved in the cytotoxic effects of lytic granules. The first electrophysiological change was a marked shortening of action potential duration (APD), occurring prior to detectable changes in the rod-shape morphology (Fig. 2). Subsequently, the resting potential was gradually reduced which, together with oscillations in the membrane potential, suggested $[\text{Ca}^{2+}]_i$ overload. These alterations culminated in a progressive reduction in the membrane potential and loss of excitability (not shown). In summary, under control conditions and after exposure to granules for 3–8 min, when morphological changes were absent or minor (slight shortening), action potential characteristics were (mean $\pm$ SEM): resting potential, -80.5 ± 0.6 mV and -73 ± 2.7 mV; action potential amplitude, 121.4 ± 1.4 mV and 102.0 ± 6.2 mV; and APD at 50% repolarization (APD_{50}), 639.4 ± 53.4 ms and 195.2 ± 51.2 ms respectively. The electrical deterioration was always associated with progressive contracture followed by myocyte destruction. It should be emphasized that in these cells dialysed with the pipette solution, the morphological changes were qualitatively indistinguishable from those occurring in neighbouring myocytes. To test whether the effects on myocyte electrophysiology and morphology are reversible, we began superfusing the bath with Tyrode's solution, after observing granule-induced reduction in APD and myocyte shortening (~ 5 min). In these experiments we found that neither the electrophysiological nor the morphological effects of lytic granules were revers-

Control



5'



6'



7'

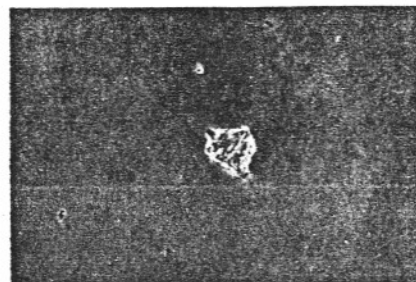


Fig. 1. Effects of lytic granules on the morphology of a guinea pig ventricular myocyte 5, 6 and 7 min after application of 7 μ l of lytic granules to the Petri dish (3.5 cm) containing the myocytes bathed in Tyrode's solution at room temperature (25°C). In numerous experiments we found that addition of granules resulted in the destruction of myocytes within 5–15 min

ible, and cell destruction advanced as under control conditions. Control experiments in which none of the above effects were seen included application of: (1) granules buffer solution (150 mM NaCl and 10 mM HEPES); (2) a post-nuclear fraction of Con A-blast prepared from murine (DBA/2 strain) spleens, which was found to be non-lytic to sheep red blood cells [2]. In the latter group ($n = 4$), action potential characteristics in the control and 15 min after exposure to the non-lytic fraction were (mean $\pm$ SEM): resting potential, -80.7 ± 0.5 mV and -78.6 ± 1.0 mV; action potential amplitude, 119.7 ± 1.6 mV and 115.8 ± 2.8 mV; and APD₅₀ 800 ± 18 ms and 806 ± 41 ms.

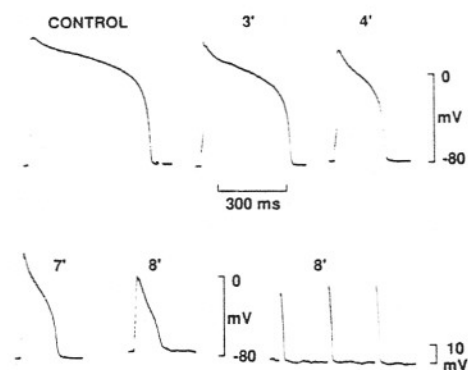


Fig. 2. Effect of lytic granules on the transmembrane action potential in a guinea pig ventricular myocyte. The action potential is shown in the control (Tyrode's solution) and at various intervals after application of lytic granules (5 μ l of granules into a 0.5 ml recording chamber). The temperature was 25.0°C, the cycle length 5 s. Bottom right panel is a recording at slower speed to illustrate the oscillations in the membrane potential

The results above illustrating progressive contracture and oscillations in the membrane potential suggest that the toxic effects of the granules are associated with $[Ca^{2+}]_i$ overload. We therefore tested whether the overload was due to Ca^{2+} release from the sarcoplasmic reticulum (SR) by treating myocytes with ryanodine (10 μ M), which blocks Ca^{2+} release from the SR. As seen in Fig. 3 and Table 1, the attenuation in the action potential occurred in the presence of ryanodine as under control conditions. Furthermore, ryanodine did not modify

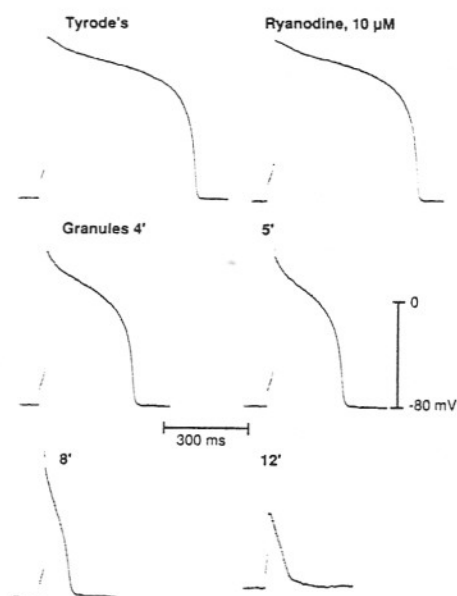


Fig. 3. Effect of lytic granules on the action potential of a guinea pig ventricular myocyte treated with 10 μ M ryanodine for 15 min prior to granule application. The temperature was 25.0°C and cycle length 5 s

Table 1. Effect of ryanodine on the lytic granule-induced alterations in the action potential in guinea pig ventricular myocytes ($n = 5$)

	Tyrode's (control)	Ryanodine (10 μ M)	Lytic granules (3–8 min)
Resting potential (mV)	-79.6 ± 0.9	-79.9 ± 0.6	$-73.6 \pm 0.5^*$
Action potential amplitude (mV)	123 ± 0.9	121.6 ± 1.2	$99.6 \pm 6.1^*$
APD ₅₀ (ms)	586 ± 75	634 ± 50	$87 \pm 38^*$

Myocytes were exposed to ryanodine for 15 min prior to granule application. APD₅₀, Action potential duration at 50% repolarization.

* Significant difference ($P < 0.01$) between lytic granules and ryanodine and control

the granules' morphological effects and cell destruction occurred in the usual sequence. We also found (not shown) that 10 mM caffeine, which depletes SR Ca^{2+} stores, did not modify the electrophysiological and morphological effects of the granules, suggesting that $[\text{Ca}^{2+}]_i$ overload does not result from Ca^{2+} release from the SR.

Hypothetically, Ca^{2+} overload could result from a granule-induced increase in the L-type Ca^{2+} current, $I_{\text{Ca,L}}$. We attempted to test this possibility by investigating the effect of lytic granules on the $I_{\text{Ca,L}}$. Commonly, $I_{\text{Ca,L}}$ is measured (in the presence of Cs^+ substituting for K^+) as the inward current remaining after eliminating the fast inward current by tetrodotoxin or by using a holding potential of -40 mV to inactivate the Na^+ channels. Figure 4A illustrates in a representative experiment the effect of lytic granules on membrane currents. Since granules markedly increased the outward current

(despite the presence of Cs^+ ions), it is impossible to quantify the direct effect on $I_{\text{Ca,L}}$. Indeed, as suggested by the effect of granules on " $I_{\text{Ca,L}}$ " current-voltage $I-V$ relations (Fig. 4B), it appears that the reduction in the inward current did not result from a "simple" block of $I_{\text{Ca,L}}$ but from an upward shift of the entire $I-V$ relations, probably due to an increase in a non-specific conductance. The only experimental evidence suggesting that $I_{\text{Ca,L}}$ is not mediating Ca^{2+} overload is that the toxic effects of lytic granules also occur in the presence of verapamil [3].

To explore the ionic mechanism(s) that might contribute to APD shortening, the $I-V$ relations of the current measured at the end of a 300 ms clamp pulse, corresponding in time to the plateau of the action potential was examined. Under normal recording conditions, this measurement reflects the inward and outward rectifying K^+ currents, I_{kl} and I_{k} respectively. In the presence of K^+ -containing Tyrode's solution (Fig. 5A), granules caused flattening of the hump which is related to I_{kl} , increased the inward and outward currents and shifted the reversal potential towards 0 mV, suggesting an increase in a non-specific conductance. The increase in the outward current, especially at positive membrane potentials, can contribute, at least in part, to the decrease in APD which will cause a negative inotropic effect. However, it should be emphasized that flattening of the outward hump and a shift of the reversal potential towards 0 mV would be expected to cause a decrease in the resting potential as well as the reduction in APD. Hence, this apparent controversy is not resolved by the present experiments and requires further investigation. Since the increase in inward and outward currents occurred after substituting K^+ (in the Tyrode's and pipette solutions) by Cs^+ (Fig. 5B), it is

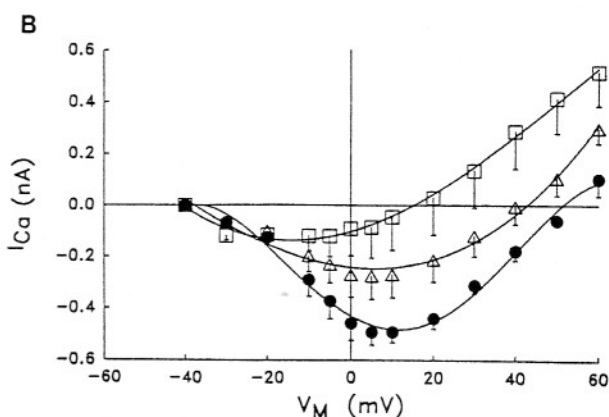
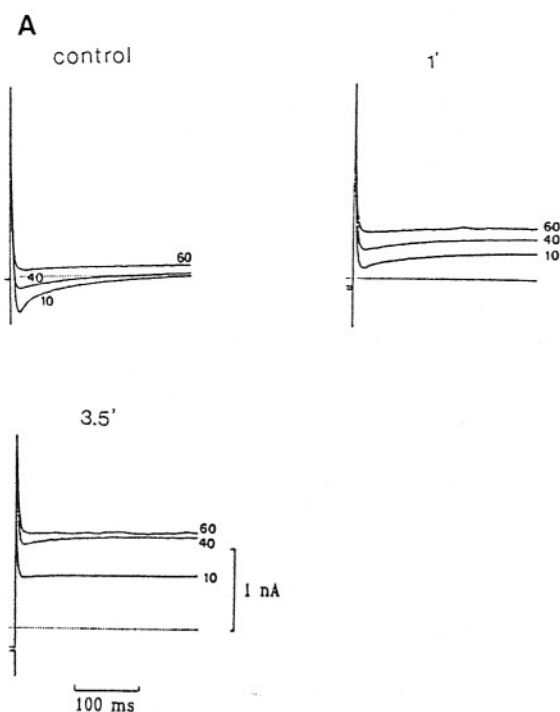


Fig. 4A,B. Effect of lytic granules on membrane currents in guinea pig ventricular myocytes. **A** Current traces were generated in response to voltage-clamp pulses from a holding potential of -40 mV to the membrane potentials indicated beside each trace. Traces were obtained during control, 1 and 3.5 min after addition of granules to the bath. **B** The effect of granules on current-voltage relations of the L-type Ca^{2+} current, $I_{\text{Ca,L}}$ ($n = 5-8$ myocytes). Results are expressed as mean $\pm$ SEM. To isolate $I_{\text{Ca,L}}$, K^+ in both the Tyrode's and the pipette solutions was replaced by Cs^+ . $I_{\text{Ca,L}}$ was measured as the peak inward (or outward) current. Symbols indicate Control ($\bullet$), 1–2 (Δ) and 3–5 ($\square$) min exposure to lytic granules

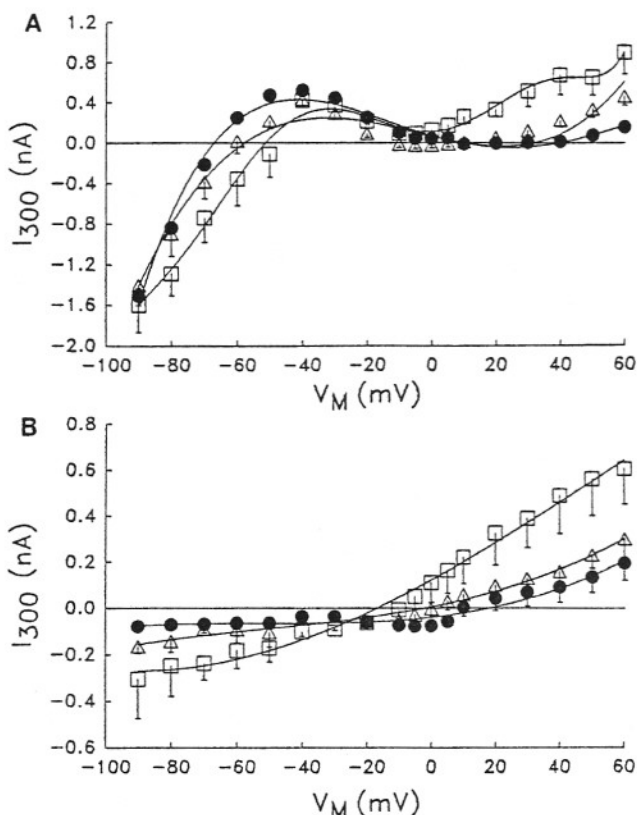


Fig. 5A,B. The effect of lytic granules on current-voltage relations ($n = 6-12$ myocytes) of the current measured at the end of a 300 ms clamp pulse, I_{300} . **A** Control (K^+ -containing) Tyrode's solution. **B** K^+ in both the Tyrode's and the pipette solutions replaced by Cs^+ . Results are expressed as mean $\pm$ SEM. Symbols indicate Control ($\bullet$), 1–2 (Δ) and 3–5 ($\square$) min exposure to lytic granules

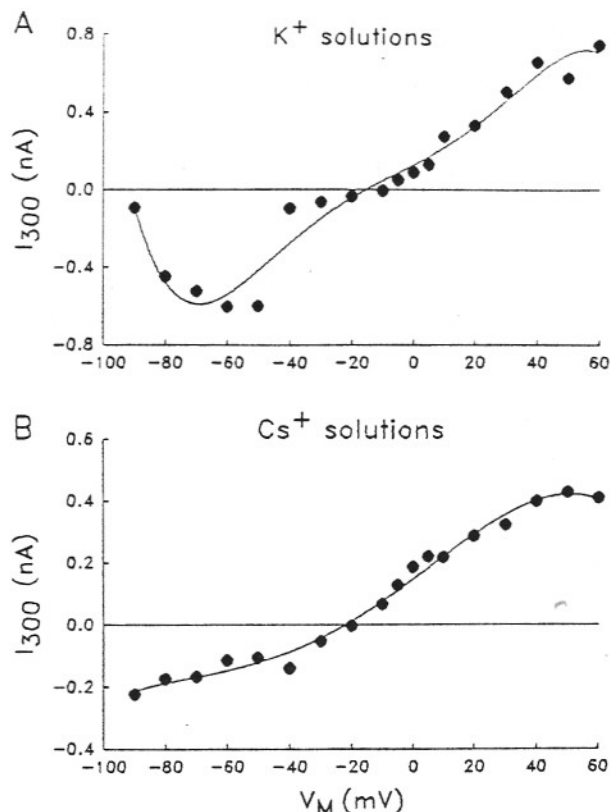


Fig. 6A,B. Lytic granule-sensitive current-voltage ($I-V$) relations of the current measured at the end of 300 ms clamp pulse, I_{300} . The curves were derived from the data shown in Fig. 5, and were constructed by subtracting the (mean) control $I-V$ relations from that in lytic granules (3–5 min). **A** Control (K^+ -containing) Tyrode's solution. **B** K^+ in both the Tyrode's and the pipette solution replaced by Cs^+

likely that lytic granules did not directly affect I_{K_1} and/or I_{K_2} . To determine the $I-V$ relations of the lytic granules-sensitive current, we plotted the difference in current between control and after 3–5 min exposure to lytic granules (Fig. 6). It is seen that, except for the most extreme negative potentials, the $I-V$ relations of the K^+ - and Cs^+ -containing solutions are quite similar; the relations appear to be linear and cross the voltage axis around -20 mV, suggesting a major contribution of non-selective ionic conductance.

As both previous reports [25, 28] and our own results suggest that cytotoxic activity is associated with $[Ca^{2+}]_i$ overload in the target cell, we used fura-2 to study the effects of granules on $[Ca^{2+}]_i$ in relation to cell length (Fig. 7). Fura-2 fluorescence images at 340 nm and 380 nm excitations (510 nm emission) and cell length (obtained from the 380 nm image) were recorded under control conditions and after exposure to granules. Figure 7A (1 of 4 experiments) demonstrates that the increase in $[Ca^{2+}]_i$ precedes detectable myocyte shortening. By 190 s $[Ca^{2+}]_i$ had risen from a basal level of about 60 nM to 175 nM, yet the cell remained almost fully relaxed. The cell reached its final stage of contracture by 515 s with an approximate $[Ca^{2+}]_i$ of 720 nM, thereafter advancing in-

to a blebbed sphere. Figure 7B demonstrates the similarity (in each cell), of the time courses of granule-induced changes in $[Ca^{2+}]_i$ and cell length. $[Ca^{2+}]_i$ increased about 2-fold from a basal level of 90.0 ± 1.5 nM, while cell length decreased less than 5% from a resting length of 99.0 ± 9.0 μ m ($n = 4$). With further $[Ca^{2+}]_i$ increase, a greater decrease in cell length occurred, followed by myocyte destruction and a large increase in $[Ca^{2+}]_i$ above 1 μ M.

Discussion

In the present study we investigated mechanisms whereby CTL damage myocardial tissue during the immunological rejection of the transplanted heart. We studied the effects of lytic granules extracted and purified from an IL-2-dependent CTL cell line. The study demonstrates that CTL-derived lytic granules induce a variety of complex changes in ventricular myocyte morphology, electrophysiology and $[Ca^{2+}]_i$. The diversity of these changes suggests that more than a single granule constituent might be contributing to the effects of whole granules, and future studies should therefore deal with effects of individual components such as perforin and serine

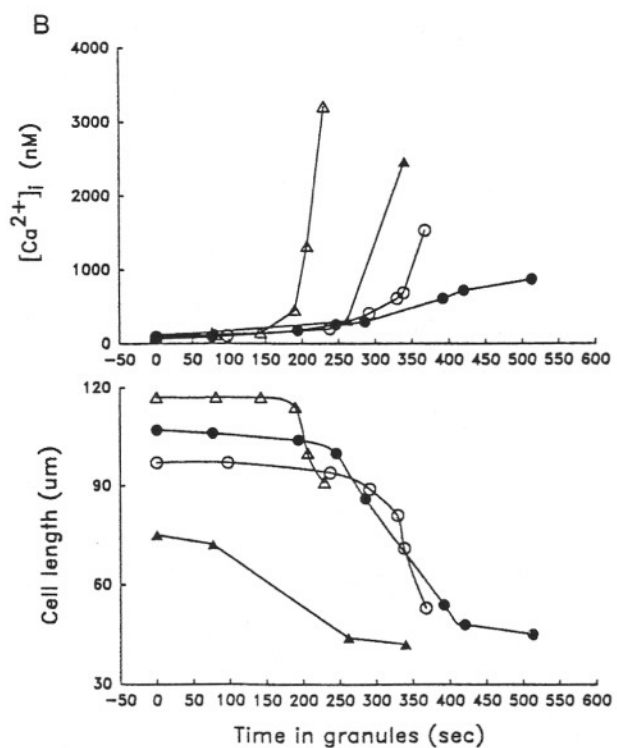
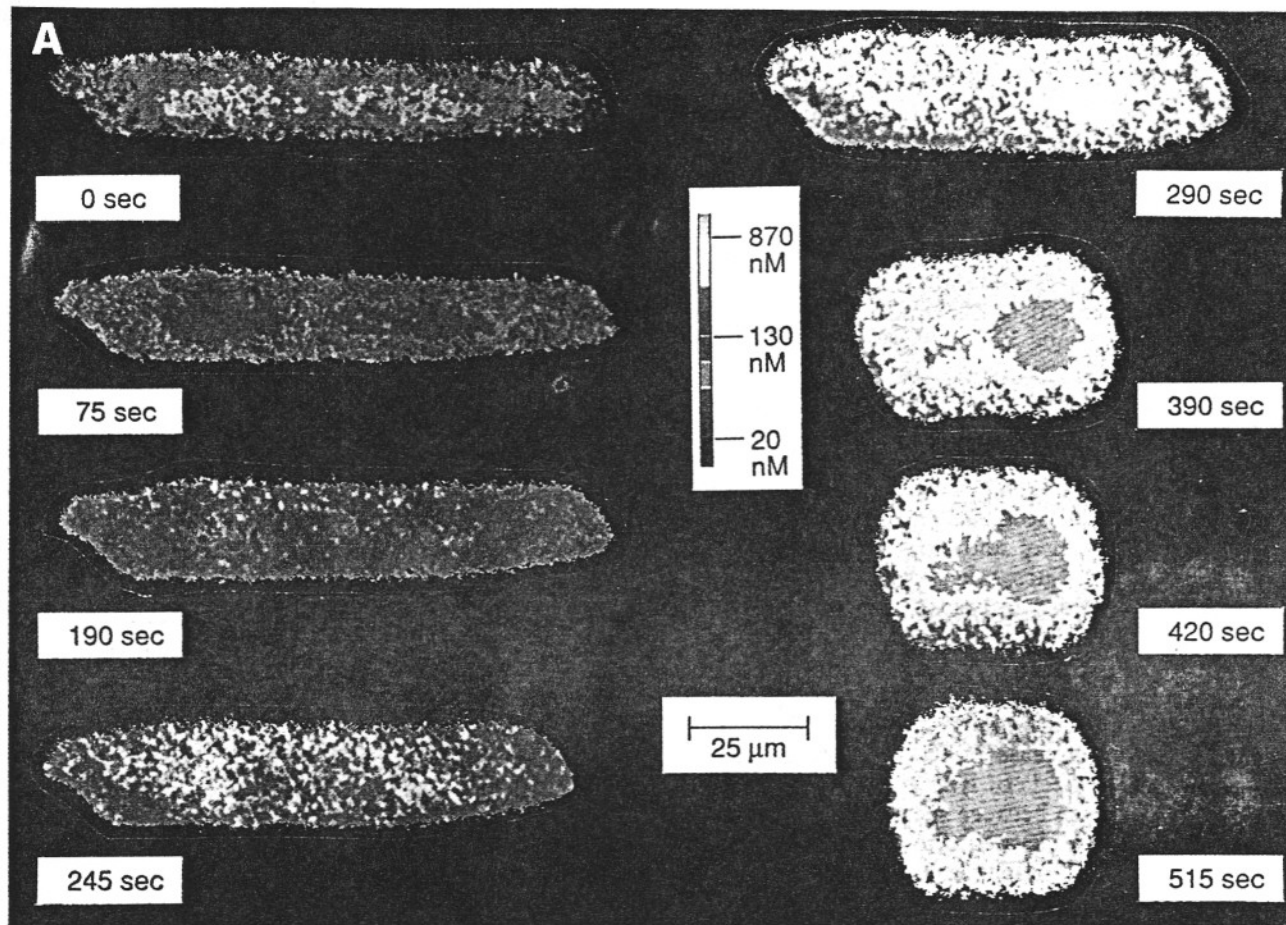


Fig. 7 A,B. Effect of lytic granules on intracellular Ca^{2+} concentration ($[Ca^{2+}]_i$) and cell length. $[Ca^{2+}]_i$ was measured using fura-2 imaging, (for details see text) **A** A representative experiment in a guinea-pig ventricular myocyte illustrating the changes in $[Ca^{2+}]_i$ and cell length induced by granules (see details in text). Initial $[Ca^{2+}]_i$ and cell length were 62 nM and 107 μ m respectively. **B** Time courses of the changes in $[Ca^{2+}]_i$ and cell length in 4 myocytes following exposure to granules. Each symbol represents data from the same cell in both panels

proteases. The alterations in action potential configuration and myocyte morphology, as demonstrated by the reduction in APD, membrane depolarization and inexcitability, and contracture and cell destruction, can be partly explained by the combined changes in $[Ca^{2+}]_i$ and membrane currents. However, as $[Ca^{2+}]_i$ was measured in resting myocytes and not in those attached to the patch pipette, the time courses of the changes in action potential and membrane currents can not be compared to that of the changes in $[Ca^{2+}]_i$. Some of the changes in action potential characteristics can be explained by the alterations in membrane currents. The reduction in APD can be accounted for, at least partly, by an increase in the outward current in the plateau range of the membrane potential. Since the increase in outward current also occurred when Cs^+ was substituted for K^+ ions in the Tyrode's solution and in the pipette, it is likely that it resulted largely from a granule-induced rise and/or induction of a non-specific conductance, and not from a specific change in I_k and/or I_{kl} . Such non-specific conductance changes have been recorded in lipid bilayers exposed to lytic granules and purified perforin [31]. In addition, the reduction in the membrane potential leading to inexcitability can be caused by the increase in the non-specific conductance with a hypothetical reversal potential close to 0 mV. The marked increase in $[Ca^{2+}]_i$ can also account for some of the effects of lytic granules, although it is difficult to distinguish between specific and non-specific consequences of an elevated cytosolic $[Ca^{2+}]_i$. For example, a rise in $[Ca^{2+}]_i$ might lead to a reduction in APD by increasing I_k [30], to oscillations in the membrane potential and to contracture and myocyte destruction. The mechanism responsible for $[Ca^{2+}]_i$ overload was not conclusively identified in this study. The ineffectiveness of ryanodine and caffeine on the granules' cytotoxic action suggests that $[Ca^{2+}]_i$ overload does not result from excessive Ca^{2+} release from the SR. Other possibilities, such as Ca^{2+} influx via a "perforin-induced channel", and/or release from other intracellular stores should be considered in future studies.

In agreement with our $[Ca^{2+}]_i$ measurements, Tirosh and Berke [29] and Poenie and colleagues [25] found that CTL/target interaction resulted in an increase in target $[Ca^{2+}]_i$ to micromolar concentrations in 250–300 s; this was mimicked by application of cytolytic granules from rat killer cells. In a related study, Hassin and co-workers [11] examined functional changes in mengovirus-infected cultured neonatal rat myocytes, induced by mengovirus-sensitized CTL. Their finding, that the attenuated contraction and impaired relaxation was associated with "oscillations in the prolonged plateau phase of the action potential", disagrees with ours. This discrepancy is probably due to major differences in the two experimental models.

The involvement of $[Ca^{2+}]_i$ overload in cell damage and death is well established and has been associated with tissue necrosis, ischaemia and immunological responses [4, 8, 15, 22, 27]. Specifically, increased $[Ca^{2+}]_i$ can cause various cell dysfunctions such as surface abnormalities, DNA fragmentation and Ca^{2+} -induced damage to mitochondria, thus affecting generation of ATP and hence the function of ATP-fuelled ion pumps [1, 7, 18,

21, 26]. Ca^{2+} activation of intracellular ATPases such as actomyosin could lead to additional ATP depletion and further suppression of $Na^+ - K^+$ ATPase [28] which would most likely cause a reduction in the resting potential.

Based on the results of the present study, and since CTL-containing lytic granules have been implicated in the process of immunological rejection of the transplanted heart (as well as other organs), it is conceivable that lytic granules contribute to the damage inflicted on the transplanted heart by the invading lymphocytes. To verify this theory we are currently investigating the electrophysiology of ventricular myocytes subjected to attack by CTL rather than by their cytotoxic granules.

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References

- Berke G, Rosen D (1988) Highly lytic in vivo primed CTL devoid of lytic granules and BLT-esterase activity acquire these constituents in the presence of growth factors upon blast transformation in vivo. *J Immunol* 141:1429–1436
- Berke G (1989) Functions and mechanisms of lysis induced by cytotoxic T lymphocytes and natural killer cells. In: Paul WE (ed) *Fundamental immunology*. Raven Press, New York, pp 735–764
- Binah O, Rubinstein I, Marom S, Berke G, Hoffman B (1990) Effect of lytic granules from cytotoxic T lymphocytes (CTL) on guinea pig ventricular myocytes. *Biophys J* 57:M–409
- Chien KR, Pfau RG, Farber JL (1979) Ischemic myocardial injury. *Am J Pathol* 97:505–529
- Copeland JG (1988) Cardiac transplantation. In: Ravitch MM, Steichen FM, Austen WG, Polk HC, Scott HW, Wells SA (eds) *Current problems in surgery*, vol 25. Year Book Medical Publishers, Chicago, pp 607–672
- Dennert G, Podack ER (1983) Cytolysis by H-2 specific T killer cells. *J Exp Med* 157:1483–1495
- Duke RC, Cohen JJ, Chervenak RJ (1986) Differences in target cell DNA fragmentation induced by mouse cytotoxic T lymphocytes and natural killer cells. *Immunology* 137:1442–1447
- Fleckenstein A, Frey M, Fleckenstein G (1983) In: Kepper D, Popper H, Bianchi L, Reuter W (eds) *Mechanisms of hepatocyte injury and death*. MTP Press, Lancaster, pp 321–335
- Gryniewicz GM, Poenie M, Tsien RY (1985) A new generation of Ca^{2+} indicators with greatly improved fluorescence properties. *J Biol Chem* 260:3440–3450
- Hamill OP, Marty A, Neher E, Sakmann B, Sigworth FJ (1981) Improved patch-clamp techniques for high resolution recording from cells and cell-free membrane patches. *Pflügers Arch* 391:85–100
- Hassin D, Fixler R, Shimoni Y, Rubinstein E, Raz S, Gotesman MS, Hassin Y (1987) Physiological changes induced in cardiac myocytes by cytotoxic T lymphocytes. *Am J Physiol* 252:C10–C16
- Hayry P, Willebrune E von (1984) The inflammatory mechanisms of allograft rejection. *Immunol Rev* 77:85–142
- Henkart PA, Millard PJ, Reynolds CW, Henkart M (1984) Cytolytic activity of purified cytoplasmic granules from cytotoxic rat LGL tumors. *Exp Med* 160:75–93
- Isenberg G, Klockner U (1982) Calcium tolerant ventricular myocytes prepared by preincubation in a "KB medium". *Pflügers Arch* 395:6–18
- Jewell SA, Bellomo G, Thor H, Orrenius S, Smith MT (1983) Bleb formation in hepatocytes during drug metabolism is caused by disturbances in thiol and calcium ion homeostasis. *Science* 217:1257–1259
- LeFrancis L, Bevan MJ (1984) A reexamination of the role of lyt-2-positive T cells in murine skin graft rejection. *J Exp Med* 159:57–67

17. Mason DW, Moris PJ (1986) Effector mechanisms in allograft rejection. *Annu Rev Immunol* 4:119-145
18. McConkey DJ, Hartzell P, Nicotera P, Orrenius S (1987) Calcium activated DNA fragmentation kills immature thymocytes. *FASEB J* 3:1843-1849
19. McKenzie IFC (1983) Allorepression. *Transplant Proc* 15:269-273
20. Millard PJ, Henkart MP, Reynolds CW, Henkart PA (1984) Purification and properties of cytoplasmic granules from cytotoxic rat LGL tumors. *J Immunol* 132:3197-3204
21. Nicotera P, Hartzell P, Davis G, Orrenius S (1986) The formation of plasma membrane blebs in hepatocytes exposed to agents that increase cytosolic Ca^{2+} is mediated by the activation of a non-lysosomal proteolytic system. *FEBS Lett* 209:1392-1394
22. Orrenius S, McConkey DJ, Bellomo G, Nicotera P (1989) Role of Ca^{2+} in toxic cell killing. *Trends Physiol Sci* 10:281-285
23. Podack ER, Dennert G (1983) Assembly of two types of tubules with putative cytolytic function by cloned natural killer cells. *Nature* 302:442-445
24. Podack ER, Konigsberg PJ (1984) Cytotoxic T cell granules. *J Exp Med* 160:695-710
25. Poenie M, Tsien RY, Schmitt-Verhulst A-M (1987) Sequential activation and lethal hit measured by $[Ca^{2+}]$ in individual cytolytic T cells and targets. *EMBO J* 6:2223-2232
26. Russel JH (1983) Internal disintegration model of cytotoxic lymphocyte-induced target damage. *Immunol Rev* 72:9-118
27. Shanne FAX, Kane AB, Young EE, Farber JL (1987) Calcium dependence of toxic cell death: a final common pathway. *Science* 206:700-702
28. Tirosh R, Berke G (1985) T-lymphocyte-mediated cytotoxicity as an excitatory process of the target. I. Evidence that the target cell may be the site of Ca^{2+} action. *Cell Immunol* 95:113-123
29. Tirosh R, Berke G (1985) Immune cytotoxicity viewed as a stimulatory process of the target. In: Henkart P, Martz E (eds) *Proceedings of the 2nd CMC Conference*. Plenum Press, New York, pp 473-492
30. Toshe N (1990) Calcium sensitive delayed rectifier potassium current in guinea pig ventricular cells. *Am J Physiol* 258: H1200-H1207
31. Young JD-E (1989) Killing of target cells by lymphocytes. *Pharmacol Rev* 69:250-314